# A Covariant Screened Theory for a Dark-Energy-Tied Galaxy Law: the Khronon Form Fails Linear Growth, an AeST Form Survives at Linear Order, and Both Fail Cluster Cores

Working note, 2026-10-22, v1.0 (draft 2). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; the theory is not closed.


**Abstract**

The owner's programme studies a gravity framework whose galactic law is $g_N = g^2/(g+a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted; its extra gravitating mass is a collisionless "cold energy" medium, not a particle. The companion note PAPER58 found that only a screened version, with a stiffness length $L_\star$, passes both Cassini and the galactic law. This note writes that screened theory covariantly in two forms and tests both. Every lane froze its criteria before computing and ran a mutation control; both $a_0$ footings ($9.36\times10^{-11}$ and $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled.

1. **Clusters fail, in both forms.** Rebuilt from the raw eRASS1 and KiDS data, the cluster cores (0.3–1 Mpc) sit 1.68–1.85 (canonical) and 1.54–1.70 (alternative) times above the law on their baryons, at $Z \approx +5$ to $+6$; any added cold profile that fixes the cores overshoots the outskirts, and AeST's own static dust has the wrong shape. This is a REAL FAILURE.
2. **v3: the Blanchet–Skordis Khronon in its literal form** (no extra curvature term, $\beta = 0$; no kinetic term for the spatial covector, $\eta = 0$; Blanchet–Skordis $K(Q)$ dust), with an independent spatial covector $\mathbf{P}$ and one stiffness term $-(\gamma_L/2)(D\cdot P)^2$. Its static limit is AQUAL with $\mu = x/(1+x)$ exactly (the free function is postulated, outside Blanchet and Skordis's admissible class). It is healthy at linear order, well posed in the weak field, and passes Cassini (all 24 cells, by $4.8\times10^7$ to $5.1\times10^9$; the frozen rule reads UNDECIDED on one float-cancellation check), the planets and the outer globulars (which set $L_\star \geq 21.68$ / 22.85 pc). **But it fails linear growth**: its free function has $J'(0) = -1$, so the Newtonian term vanishes on FRW, and the dust mass $\mu_K$ has no allowed window (a gap of 24–172×).
3. **v4: AeST with the same stiffness term and C1's capped free function** keeps everything v3 passed (the exact law, Cassini, BBN) and removes the growth failure at linear order: the Newtonian term survives in the infrared for any free function. Its one flagged health defect was a mis-specified criterion (it flags ordinary Newtonian dust); the mode is ghost-free **at linear order only**, with a bounded growth rate. Open: a contact-term binding up to $2.3\times10^4$ times Newton between $\mu$ and $k_c$, the stiffness term's lack of a causal cone, nonlinear behaviour and strong fields. v4 is the better-surviving candidate; it is not established.
4. **Other tests.** Outer galaxy lensing: UNDECIDABLE (the frozen REAL FAILURE is not robust to the host veto). Dwarf satellites: a +0.89 dex tension, supply only conditionally met. Binary pulsars and GW170817: no constraint. ALMA: expected unscorable. $\kappa$, $L_\star$ and the dust abundance are fitted, not derived.
5. **Pending, stated plainly:** the native CMB (validated code, no native spectrum run; it can test only the dust sector and the cosmological $G$ ratio, because $a_0$ is absent at linear order) and the native cosmic-shear run (running, no verdict). Gaia DR4 is the sealed decisive test of screening: screened $\gamma_v - 1 \approx 8.1\times10^{-6}$ against $\gamma_v \approx 1.17$ for the unscreened reading.


## 1 Setting and rules

- $\kappa = \frac{1}{2}$ is fitted. The thermal value $\kappa = Z/4\pi$ is Milgrom's relation $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ (arXiv:1605.07458, eq. 5), a postulate here (PAPER58). Footings: canonical $a_0 = 9.36\times10^{-11}$, alternative $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled.
- Status words follow the record: DERIVED (from listed premises), COMPUTED, POSTULATED, FITTED, OPEN, PENDING. The source document is `covariant_push_2026_10_22/THEORY_V3_DRAFT.md` (commit 5d7308f3a2); every claim below cites its lane and commit. Lanes C1–C8 are in `covariant_push_2026_10_22/`; D10 and E6 in `doors_push_2026_10_22/`.
- **Two forms.** v3 is the Khronon form of the record's theory draft; v4 is the AeST form found in the rebuild lanes (`rebuild_2026_10_22/`, F1–H3), added to the draft as its §1b (commit 8bcafea2fa). The rebuild lanes were rerun by the coordinator; F1 and H2 were reproduced on an independent KiDS extraction.
- Table 1 lists the lanes used here that are not in PAPER58.

Table 1: Lanes of the covariant push, two late doors lanes and the rebuild lanes. Commits: criteria / result. [a]Frozen rule REAL FAILURE; not robust, so UNDECIDABLE under a post-result amendment.

| lane | question | verdict | checks | commits |
|---|---|---|---|---|
| C1 | AeST embedding | EMBEDS WITH EXTRA TERM | 8/8 | d9a9a17f4e / 6502ef90a9 |
| C2 | AeST cosmology, native $H_0$ | COMPUTED + DATA-GATED | 9/9 | ac4b30ff62 / 2dd388643a |
| C3 | AeST scales vs $L_\star$ | SCOPED NO-GO | 9/11 | c16ef33ae5 / fd7196584c |
| C4 | Khronon parent | EMBEDS WITH EXTRA TERM | 8/8 | b0ea16ecf1 / 5f4f845554 |
| C5 | health | CONDITIONALLY HEALTHY | 7/10 | f002ce909a / f3e7d4d661 |
| C6 | Cassini, literal form | UNDECIDED (frozen) | 7/8 | 1b2ced55a4 / ae428dc252 |
| C7 | native CMB code | RUNNABLE NOW | 7/7 | milestones to b0ced915c8 |
| C8 | instantaneous sector | WELL-POSED + UNCONSTRAINED | 8/8 | f2399aaf5e / aa113a5e96 |
| D10 | native $L_\star$ floor | CRISP FINDING | 10/10 | e3333cb274, A1 ecec1ce9e7 / 3ac0573063 |
| E6 | native catchment | CRISP FINDING | 5/5 | c42925d9dc, A1 5c0669a33d / 0bea588820 |
| F1 | cluster cores, rebuilt | REAL FAILURE | 6/6 | 9fd5aee16e / f1803b0256 |
| F2 | outer lensing, rebuilt | UNDECIDABLE[a] | 9/9 | e9816b47d2 / 7c8c3d5302 |
| F3 | growth vs v3 | v3 growth FAILS | 9/9 | 0bef684eea / 059766caa5 |
| G1 | old engine's excess | ARTEFACT (old engine) | 7/7 | 9921a5172a / a048e6d468 |
| G2 | pre-infall retention | FAILS (native) | 4/6 | 92716d0148, A1 b3fbb9ed9d / 3cdcf3b4c5 |
| H1 | AeST free function | NONE IN THE FAMILY | 6/6 | a4d76f492e / 1654b84f78 |
| H2 | AeST cluster mass | CANNOT | 6/6 | 364a56c250, A1 5160a89939 / 385f61a14c |
| H3 | the health criterion | CLAIM FAILS (no ghost) | 4/5 | 04355110c4 / 6b905cb6d8 |

## 2 The v3 action (C1, C3, C4, C5)

The Blanchet–Skordis Khronon action is $S = (c^3/16\pi G)\int \sqrt{-g}\,[R - 2J(Y) + 2K(Q)]$, with a preferred foliation $\tau$, khronon acceleration $A_\mu$ and $Y = A\cdot A/c^4$ (Blanchet & Skordis 2024, 2025; the equations were read via the arXiv HTML and re-derived symbolically in C4, not verified on disk). The screened theory is:

- **Literal form.** $\beta = 0$ (no extra curvature term), $\eta = 0$ (no kinetic term for $\mathbf{P}$), Blanchet–Skordis $K(Q)$ dust.
- **The free function.** $J = J_x = \Lambda - 2a_0^2[y - \ln(1+y)]$, $y = \sqrt{Y}/a_0$, gives $1 + J' = y/(1+y)$: AQUAL with $\mu = x/(1+x)$ exactly (C4, sympy). But $J_x/\sqrt{Y} \to -2a_0$ at large $Y$, so it lies outside Blanchet and Skordis's admissible class, which demands $J \to$ const. POSTULATED.
- **The addition.** $\mathbf{P}$ promoted to an independent spatial covector ($n\cdot P = 0$) with one stiffness term $-(\gamma_L/2)(D_iP^i)^2$, $\gamma_L = KL_\star^2/2$: one new field and one new constant. The pure Khronon cannot screen the Sun: with $J_x$ the planets see the full $a_0$, $1.08\times10^4$–$8.07\times10^4$ times their bounds; with a turn-off $J$ the AQUAL $Q_2$ still fails by at least 2.44× (canonical) and 2.85× (alternative) (C4, b0ea16ecf1 / 5f4f845554).
- **Static limit.** $\phi = \psi$ (lensing sees the dynamical potential) and $\nabla\cdot[(1+J')\nabla\psi] = 4\pi G\rho$ for any mass distribution: the field equation is AQUAL, not QUMOND, matching PAPER58's E2. The curl that AQUAL needs lives in the unconstrained auxiliary $P^*$ ($|\nabla\times P^*|/|\nabla P^*| = 0.126$ on a two-mass test), not in the foliation.
- **Relation to AeST (C1, C3).** AeST (Skordis & Złośnik 2021) reproduces AQUAL with $\mu = x/(1+x)$, the clock term and the curl half of the stiffness exactly, but the longitudinal stiffness $\gamma_L(D\cdot P)^2$ is identically zero in AeST and in any first-derivative extension; it needs second derivatives of AeST's scalar (C1, d9a9a17f4e / 6502ef90a9). No term in AeST's class screens small systems (C3, SCOPED NO-GO, c16ef33ae5 / fd7196584c): AeST's mass term acts at large $r$ and in the opposite direction. So $L_\star$ is a new constant of the covariant theory.
- **Relation to DSF-21's A33.** A33 is this theory with $\eta > 0$. With the Khronon's curvature term ($\beta = 0$), $\eta > 0$ gives a gradient instability wherever the field is nonzero; in the Solar System at $k = 1/\mathrm{AU}$ the rate is 83.3 /yr (canonical) and 95.09 /yr (alternative). So $\eta$ is dropped (C5).
- **A closure identity (C4).** If $J(\infty) = 0$ (Blanchet–Skordis 2025), the action gives $\Lambda c^4/a_0^2 = 2F(y_t)$, twice the record's closure, so the turn-off scale would be 33.94 (canonical), 23.70 (alternative) or 33.98 ($\kappa = \frac{1}{2}$). This applies only to the turn-off branch, which fails Cassini anyway, and it does not select $\kappa$.

# 3 Health (C5)

Criteria f002ce909a, result f3e7d4d661. Verdict CONDITIONALLY HEALTHY, and only in the literal form.

- **The literal form.** No ghost and no gradient instability on every background tested where the frozen-coefficient (WKB) expansion is valid; only the two tensor modes propagate, at $c$; **P** and the khronon scalar obey elliptic constraints; the $\omega = 0$ mode has positive energy at every $k$; the FRW dust has $c_s^2 = (\bar{Q} - 1)/\bar{Q}$. Conditions: $\gamma_L > 0$; $W$ convex (the record's $W$ meets this at every field strength); a positive dust charge; $\beta = 0$ and $\eta = 0$ exactly.
- **What fails.** $\eta > 0$ with $\beta = 0$: ill-posed (above; the stability boundary is the record's $\beta > 2\eta a_0^2/K$, confirmed at $\beta = 1.48707 \times 10^{-19}$). $\beta > 0$ with the $K(Q)$ dust: an infrared gradient band. AeST plus the stiffness term was ranked second: its $\omega = 0$ mode has negative energy at every $k$ in vacuum, and the stiffness term gives an unbounded speed. (H3 later showed the first defect is the energy criterion flagging ordinary dust; Section 6.)
- **Checks.** 7/10, exit 1. Three frozen checks fail and are kept: H5 (a weak-field factor fails at neutron-star compactness), H6 (all failing points lie where the background is not local; this reading is post hoc) and H9 (the $\beta > 0$ dust band, physics). Strong coupling is far below every sub-Hubble scale.

# 4 Well-posedness and causality (C8)

Criteria f2399aaf5e, result aa113a5e96. WELL-POSED + UNCONSTRAINED, scoped to the weak-field leaf problem; 8/8 scored checks.

- **Well posed.** On each leaf the **P** and lapse equations are the Euler–Lagrange equations of one strictly convex functional, so the solution exists and is unique on a bounded domain for every source below the cap ($\gamma_L > 0$). Because the action has no $\dot{N}$ or $\dot{P}$, the time-dependent leaf problem is exactly the static one. OPEN: the infinite-volume theorem, the coupled system with GR's tensor sector and the dust, and strong fields.
- **Criterion A is violated; criterion B is met.** The response is a real, gauge-invariant infinite-speed influence on the khronon's leaves. It never goes backward in $\tau$, so the single-global-time criterion holds. Its size: moving 1000 kg by 1 m changes the force 1 km away by $1.2 \times 10^{-46}\,\mathrm{m\,s^{-2}}$ at $L_\star = 24\,\mathrm{pc}$.
- **No existing data constrain it.** Binary pulsars: no radiative channel, and the conservative correction is at most $3.2 \times 10^{-14}$ of the double pulsar's precision. Preferred frame: an $\alpha_2$-equivalent of $1.7 \times 10^{-7}$ of the bound (this also corrects the record's A22 D5, which missed the leaf tilt). GW170817: $\Delta v/c \leq 6.2 \times 10^{-32}$.

# 5 Linear growth: v3 fails (F3, G1)

Criteria 0bef684eea, result 059766caa5; 9/9.

- **v3 fails linear growth as written.** With $J_x$, $J'(0) = -1$ exactly, so the Newtonian term vanishes on FRW; growth like GR on dust holds only below the dust's mass scale $\mu_K$. To keep $\sigma_8$ within 5% of the dust limit $\mu_K$ must be at least 361–2542 $h$/Mpc, while Blanchet and Skordis's bound caps it at 14.8 $h$/Mpc: a gap of 24–172×. Inside the allowed range linear $\sigma_8$ is only 0.04–0.39 of the dust limit; above it the dust forms galaxy halos and double-counts against the law. Whether the cubic deep-MOND term restores growth needs a native simulation.
- **The record's growth failures belong to the old settling rules,** not to v3. The KiDS $\Delta\chi^2 = +68.7$ (CFG596) is reproduced exactly (+68.73) and comes from the settling rules and the QUMOND phantom, which v3 does not contain. The old engine's 15–19% power excess is not even converged: it grows with the rule's grid, 0.040 / 0.077 / 0.163 at $128^3$ / $256^3$ / $512^3$, and a smooth cap makes it worse (G1).
- **In the $\mu_K \to \infty$ limit** (the dust limit), linear $S_8 = 0.845$ / 0.814 against raw KiDS on $k \leq 0.3\,h$/Mpc ($0.754 \pm 0.063$ / $0.761 \pm 0.060$): CONSISTENT. The nonlinear part needs a native simulation.

# 6 v4: AeST with the stiffness term and a capped free function (C1, C5, H1, H3)

- **The action.** Skordis–Złośnik AeST with C1's capped free function (its static limit is the canonical law exactly) plus the same single stiffness term $\gamma_L$, which AeST lacks (C1). POSTULATED.

- **What it passes.** Growth: AeST's Newtonian term survives in the infrared for any free function ($\Gamma^2(k \to 0) = 4\pi\tilde{G}\bar{\rho}$), unlike v3's Khronon. The galactic law exactly; Cassini (screened, as v3); BBN ($\lambda_s = \infty$). No member of Skordis–Złośnik's own family passes all of these (H1, a4d76f492e / 1654b84f78): BBN needs $\lambda_s \geq 24.7$, where their interpolants lose to the canonical law on SPARC at $z = -2.37$ to $-2.74$ (canonical) and $-2.78$ to $-3.64$ (alternative). Only the capped function gives the law, and it failed only C5's health criterion.
- **That health criterion was mis-specified (H3, 04355110c4 / 6b905cb6d8).** The same energy test flags ordinary self-gravitating Newtonian dust and GR plus k-essence dust. The AeST mode is non-propagating dust, ghost-free at linear order on the vacuum, with growth rate bounded by $\sqrt{4\pi\tilde{G}\bar{\rho}}$ (maximum 0.99991 of it) and a band that ends at $k_c \approx 243\,\mu$. The frozen claim "only the Jeans instability" FAILS on three lines, none of them a ghost.
- **Open for v4.** Between $\mu$ and $k_c$ the binding is AeST's contact-term static law, up to $2.3 \times 10^4$ times Newton near $k \approx 185\,\mu$; its consequences for small-scale structure are OPEN. C5's second AeST defect (the stiffness term gives no causal cone) is untested. A linear FRW run with C7's code at this free function, nonlinear behaviour and strong fields are OPEN. So v4's health holds at linear order only.

# 7 Parameters

- $a_0$ **($\kappa$): FITTED.** $\kappa_\Lambda = 0.417 \pm 0.095$ (CFG493); $0.414 \pm 0.094$ on a $\Lambda$CDM-free background (PAPER58, E4).
- $L_\star$**: FITTED, a new constant, lower bound only.** With the theory's own field equation in the Galactic field (D10, result 3ac0573063), the floor is $L_{\star,95} = 21.68\,\mathrm{pc}$ (canonical) / $22.85\,\mathrm{pc}$ (alternative) with the stellar-population $M/L$; over the $M/L$ priors and the Milky-Way bracket it spans 2.93–22.67 / 4.40–23.80 pc and straddles 20 pc. The record's upper edge is about 90 pc (PAPER58, D8).
- **The dust abundance: FITTED.** It is an initial condition, the khronon's conserved shift charge (C4) or AeST's (C2); nothing in the action fixes it.
- $G_{\mathrm{cosmo}}/G_N$**: bounded.** BBN (the record's A78): $G_{\mathrm{cosmo}}/G_N > 0.9611$. A Planck-lite emulation gives $\sigma = 0.027$ (C2, wave 2, 5e93769089). In AeST it can only be lowered, and the same BBN line gives $K_B < 0.0778$ (C2).
- $\mu_K$ **(v3's dust mass): no allowed value** (F3, above).
- $u$ **(the retained fraction):** a measured, convention-dependent function of scale with no mechanism (PAPER58, D6, E1, E4).

# 8 Tests

## 8.1 Galaxies

The static limit is the law itself (AQUAL, $\mu = x/(1+x)$; C4, PAPER58 E2). Galaxy rotation curves therefore test the law, as in the companion notes; this theory adds nothing at galactic scales.

## 8.2 Cassini and the planets (C6, D7d)

Criteria 1b2ced55a4, result ae428dc252.

- **The literal form passes Cassini.** All 24 cells ($L_\star = 24$, 50, 90 pc; both footings; four Galactic fields) pass Hees et al. (2014) and Park et al. (2026). At 24 pc $Q_2 = 0.93$–$1.86 \times 10^{-34}\,\mathrm{s}^{-2}$; at 90 pc $1.76$–$3.52 \times 10^{-36}$. The margin below the Hees line is $4.8 \times 10^7$ to $5.1 \times 10^9$.
- **Mechanism.** Inside $L_\star$ the Sun's field is a constant hedgehog $GM/L_\star^2\,\hat{r}$ with no quadrupole. Removing the curl stiffness (D7d used it) lowers $Q_2$ by 9–19%; it does not open a quadrupole.
- **The frozen rule reads UNDECIDED.** One validation check (C2) fails at $3.8 \times 10^{-11}$ against a $10^{-12}$ line: rounding in $(\kappa^2/2 + 1) - 1$ at $\kappa = 0.01$. The identity itself is exact in sympy. The script exits 1 (7/8).
- **Planets.** The anomaly is the hedgehog $GM/L_\star^2$: at 24 pc Mars is at 0.173 of its bound, Saturn 0.035 and Earth 0.028.
- **The unscreened law, for contrast,** fails Cassini by 2.44–5.19× under AQUAL (PAPER58, E2).

### 8.3 Cluster cores (F1, H2): a real failure for both forms

F1 (9fd5aee16e / f1803b0256) rebuilt the eRASS1 × KiDS cluster-core stack from the raw data with its own KiDS reader and estimator (326 clusters; it reproduces the record's stack within $1\sigma$ in every bin).

- **The law on the clusters' baryons fails.** In the primary rebuild (AQUAL, native $h$, gas rescaled, miscentring) $A_{\rm MID} = 1.68$–$1.72$ (canonical, $Z = +5.6$ to $+6.2$) and 1.54–1.58 (alternative, $Z = +4.9$ to $+5.5$); with the boost sign fixed (post-result) 1.81–1.85 / 1.66–1.70. The field equation, native $h$, gas calibration, $R_{500}$ and miscentring each move it by at most 4%. At the corner most favourable to the law it is still 1.33–1.37 (canonical) and 1.22–1.26 (alternative), and that corner pushes the outskirts below the law.
- **No added cold profile fixes it:** any profile that fixes MID overshoots the outskirts. The theory's dust inside 1 Mpc is not derived.
- **AeST's static dust cannot supply it (H2, 364a56c250 / 385f61a14c):** its mass term ramps across 0.3–1 Mpc and oscillates negative outside, where the data need a flat excess then none. Its best profile $\chi^2$ is 27.7–39.6, against 27.3–41.8 for the law alone and a pass line of 11.07. The frozen rule's one-sided pass was corrected post-result to CANNOT. A cold-dark-matter-like branch of AeST's dust would need a native AeST N-body code.

### 8.4 Outer galaxy lensing (F2): undecidable

F2 (e9816b47d2 / 7c8c3d5302) rebuilt the isolated-galaxy outer-lensing test with the AQUAL equation solved exactly. The frozen rule reads REAL FAILURE, but a post-result diagnostic shows the frozen sample leaks satellites' hosts, so the reading is UNDECIDABLE. The record's $\Delta\chi^2 = +100.1$ came from a catchment supply edge the covariant theory lacks; without it the far-outer amplitude falls from 2.45 to 0.96 of the data (canonical). Any measured external field makes the theory's outer prediction far too small: the weakest on-disk field gives an amplitude of 4.45, and the sample allows $y_e \lesssim 3 \times 10^{-5}$. The isolated stack's outer signal moves by $0.74 \pm 0.09$ ($8.6\sigma$) with the isolation cut.

### 8.5 Outer globular clusters (D10)

Each cluster was solved as a Plummer sphere in the Galactic field with the literal form's own equations. The unscreened law ($L_\star = 0$) is excluded in all 18 cells ($\Delta\chi^2 = 4.9$–$22.8$), and no finite $L_\star$ beats Newton. So the globulars give a lower bound only. Pal 4 and Pal 14 dominate; the bound is limited by the stellar $M/L$ of Pal 4.

### 8.6 Dwarf satellites (E3, E6)

- The satellites sit $+0.89 \pm 0.13$ dex above the screened prediction (PAPER58, E3).
- **A pre-infall retention rule does not close it (G2, 92716d0148 / 3cdcf3b4c5):** on native inputs the offset stays $+0.92 \pm 0.14$ dex (canonical) / $+0.89 \pm 0.15$ (alternative), because the retained cold energy pays the law's phantom first. It closes only with a ΛCDM subhalo mass put in by hand, and then overshoots 8 systems.
- **The framework's dynamics cannot fix a satellite's catchment** (E6, c42925d9dc / 0bea588820): it is set by the patch's initial overdensity and infall epoch. Median $z_{\rm req} = 11.28$; the required catchment is an ordinary $1.21\sigma$ fluctuation in an externally calibrated initial field.
- **The satellites' own chemistry** (leaky box with their [Fe/H], no ΛCDM input) bounds the catchment from below at a median 38.33 times the present baryons, so the median need is 0.211 of the supply; 6 of 36 tested systems are short at central values and none at $2\sigma$. This is CONDITIONAL on $\zeta = 1$ winds and no pre-enrichment, and it still needs a mechanism that packs the cold mass inside the half-light radius, which the record does not have.

### 8.7 Binary pulsars and gravitational waves (C8)

See Section 4: no constraint; tensor speed exactly $c$.

### 8.8 ALMA (D4)

Expected unscorable on both dates (gate S/N 0.04–0.58 against a required 5). The plans are sealed; addendum E (ffb14e40bc) adds the $n = 1$ and $n = 2$ kernels as labelled parallel readings beside the $\nu_{\rm mono}$ verdict kernel, and the harness has the kernel switch (92fa9f9471).

## 9 Sealed predictions for data still to come

- **Gaia DR4 wide binaries (20 kAU).** Screened: $\gamma_v - 1 = 8.1 \times 10^{-6}$ / $1.9 \times 10^{-6}$ / $5.8 \times 10^{-7}$ at $L_\star = 24$ / 50 / 90 pc (C6). Unscreened law-first (AQUAL): 1.167 (canonical) / 1.199 (alternative) at the primary external field. Scored on the whole-catalogue $\hat{\gamma}$ under the owner's pre-data $\sigma_{\rm sys}$ rule (pre-DR4 note amendment A3, 3d82af0a7e): definition (b) primary, (a), (e), (g2) must agree, else UNDECIDED; footings scored separately. Expected power at 30,000 pairs under (b), from DR3 (non-scoring): 3.21–3.96$\sigma$ canonical.
- **ALMA U4** (public 2026-11-06) and **GS4** (2026-11-28): frozen CFG806/CFG809 plans plus addendum E; expected unscorable.

## 10 Pending

- **Native CMB: pending.** Lane C7 patched CLASS v3.3.4 with AeST's linear equations, derived from the action; the patched code reproduces pristine CLASS in the exact dust limit to $2.2 \times 10^{-6}$ and passes its validation (7/7, milestone 4, b0ced915c8). No native spectrum has been run or confronted with data. When it is, it can test only the dust sector and $G_{\rm cosmo}/G_N$: $a_0$ is absent from the background and from linear order (C2: the invariant that carries $a_0$ vanishes on FRW at first order), so the CMB cannot test the galactic law.
- **Native cosmic shear: pending.** The record's canonical settling rules are excluded by raw KiDS (CFG596, $\Delta\chi^2 = +68.7$). The pre-registered native $1024^3$ rerun (CFG960) was restarted under amendment A6 (f72bef68ea; a bit-identical lower-memory engine) and was running past step 90 of 150 when this note was written; it has no verdict on either footing.

## 11 What this is not

- **Cluster cores are a real failure** of both forms (F1, H2).
- v3 fails linear growth (F3); v4's health holds at linear order only, with its contact-term binding and causal cone open.
- Not a UV completion: an effective theory with a preferred foliation.
- Not derived constants: $\kappa$, $L_\star$ and the dust abundance are fitted; the free function $J_x$ and the stiffness term are postulated.
- The CMB cannot test the galactic law at linear order.
- Criterion A (no influence faster than light) is violated, by a real signal too small for any test identified.
- The dwarf supply is conditional (chemistry assumptions, no packing mechanism); the dwarfs remain in tension. Outer galaxy lensing is undecidable.
- No particle species is added: the extra mass is the cold-energy sector, and the dust here is a field sector. No result here is evidence for the framework over its rivals.

**Disclosures.** C3 scores 9/11 with two frozen fails kept and a post-result amendment (6dba29d773). C5's three frozen fails (H5, H6, H9) are kept; its first complete run (6/10) had two code bugs, fixed with no pass line changed. C6's frozen verdict is UNDECIDED (float cancellation); its earlier executions and a 1.04 GB mutation run are disclosed. C8's first run (6/8) failed on rounding and a solver limit, and peaked at 2.3 GB; fixes changed no pass line. D10's amendment A1 is pre-result. E6's amendment A1 is pre-result and A2 post-result. The Khronon equations come from the arXiv HTML via the C4 lane and are not verified on disk; the AeST sources are on disk (C2 provenance). Rebuild lanes: F2's frozen REAL FAILURE is kept beside its post-result UNDECIDABLE reading; G2 scores 4/6 (two MC-comparison fails kept); H2's frozen one-sided rule was corrected post-result; H3's frozen claim fails on three lines and its K5 fail is kept; F3's amendment A1 is post-result. The pre-DR4 note's amendment A3 repeats a statement about the alternative footing that PAPER58 v1.1 withdrew; the scoring rule does not depend on it.

Repository: `covariant_push_2026_10_22/` (C1–C8, `THEORY_V3_DRAFT.md`); `doors_push_2026_10_22/` (D10, E6; D2, D4, D6, D7d, E1–E4 via PAPER58); `prep_2026/gaia_dr4_prep/` (pre-DR4 note, A1–A3). Companion notes: PAPER58 v1.1 (doi:10.5281/zenodo.23579379), PAPER57 (doi:10.5281/zenodo.23532885), PAPER49 v1.2 (doi:10.5281/zenodo.23533323). Theory: L. Blanchet, C. Skordis, JCAP 11 (2024) 040 (arXiv:2404.06584); L. Blanchet, C. Skordis, arXiv:2507.00912 (2025); C. Skordis, T. Złośnik, PRL 127, 161302 (2021) (arXiv:2007.00082); M. Milgrom, arXiv:1605.07458. Bounds: A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, PRD 89, 102002 (2014); R. S. Park et al., PRD 114, 024066 (2026). AI assistance: Anthropic's Claude (lanes, review, drafting).